FOR INVESTORS — THE ECONOMICS, AT DOSSIER STANDARD

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

Tunisia, audited — investor economics

The FX bridge by leg, per-pathway DSCR computed in code, the capital stack, and the exact signatures that move “commercially validated” from zero.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The FX bridge, tiered by confidence

LEG	$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	$75-86M	**STRONG**
Solar fuel substitution (500 MW)	$95-116M	**PLAUSIBLE**
Hydrogen → ammonia (loop-internal offtake)	$7-25M	*SPECULATIVE*
Terfas (desert truffles)	$14-29M	**PLAUSIBLE**
Food import substitution	$30-30M	*SPECULATIVE*
Compute colocation services	$15-15M	*SPECULATIVE*
ESG financing delta	$8-12M	**PLAUSIBLE**
Total (mid)	**~$279M/yr**	**tiered by grade below**

$80M carries a proven flowsheet or a cited anchor; $137M is gated; $61M rests on assumed volume or price. Zero dollars are commercially validated — the first buyer moves the ledger.

1.1 The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **$9.34B** = 4.3 months of imports; food $3.00B + fuel $4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food ($3.00B) and fuel ($4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop’s answer: owned export engines earn ~$117M/yr of gross exports outside the Fed cycle; the substitution legs take $152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 →

BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

The macro context is the thesis: the same Fed transmission that sets the emerging-market cycle prices this state's debt, and the loop's engines are the only line item in the country that earns hard currency *outside* that channel. That is why the FX bridge above is the investor's first table, not an appendix.

2. Pathway economics — computed in code (nations_v5.py)

P2 · Strategic metals (acid-stream U + heavy REE)

LINE	VALUE
Revenue	\$75-86M/yr (716.5k lbs U&sb3;O&sb8; @ \$80-95/lb + ~585 t REO)
OPEX	\$54-74M/yr (SX \$59-83/lb + shared circuit)
Gross margin	0-37%
CAPEX / payback	~\$100M / 6.3 yr (mid)
DSCR (60% debt @7%/15y)	2.40 — passes at contract prices
Gate	pilot U recovery ≥85% · all-in ≤\$75/lb · offtake signed

P3 · Solar IPP (500 MW) — the honest finding

LINE	VALUE
Revenue / OPEX	\$27.2M / \$8M (500 MW × 20% CF × \$31/MWh)
CAPEX	~\$550M
DSCR @8%/20y	**0.46** — fails
DSCR @4.5% DFI/25y	**0.69** — fails
Bankable region (DSCR ≥1.15)	\$0.70/W + 21% CF · \$0.70/W + 22% CF · \$0.70/W + 23% CF · \$0.70/W + 24% CF · \$0.80/W + 24% CF
Clearing PPA	≥\$45652/MWh at mid-case capex/CF
Verdict	bankable ONLY lean, inside the region, or with grant/sovereign support — published, not hidden

P4 · Municipal desalination (150,000 m^3/d)

LINE	VALUE
Revenue	\$30.1M @\$0.55 → \$38.3M @\$0.70 blended
OPEX / CAPEX	\$24.6M (@\$0.45/m^3) / ~\$165M
DSCR @\$0.55	**0.54** — fails
DSCR @\$0.70	**1.36** — clears

LINE	VALUE
Gate	≥60% contracted offtake · blended ≥\$0.68/m^3 · brine permit

The water stack — line-by-line, with each credit's failure mode

The \$0.347/m^3 audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

LINE	\$/M^3	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m^3 at the loop's own PPA + capex + O&M	\$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to \$0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)	−\$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs \$0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	−\$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin
Brine valorization — salt/Mg evaporated on the loop's own free heat, sold at the park gate (of-take signed; nothing hauled)	−\$0.137	DERIVED — brine is the desal's own by-product	if the of-take is not signed by FID, the credit degrades to −\$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−\$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −\$0.016
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−\$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−\$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−\$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	**\$0.347/m^3**	**28% of SONEDE's \$1.00-1.50 (Pg16)**	**the degraded cascade: if every soft credit fails, water still lands at \$0.521/m^3 — under SONEDE's own \$1.00–1.50, and the \$0.68 gate still clears with margin. The credits are the margin, not the plan.**

The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at $0.21 is 7× the loop's $0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m^3/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past Arthrospira's ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Solar electrons	500 MW fleet	desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher	876 GWh/yr
SWRO permeate	coastal desal	city offtake · electrolyzer feed · ponds · greenhouses	150,000 m^3/d
SWRO brine	coastal desal	salt + magnesium valorization, evaporated on the loop's own free heat	122,727 m^3/d → 58 kt Mg/yr
Brine bittern	valorization	Mg(OH)&sb2; fire retardant → **the board's own chemistry**	140 kt/yr = 140% of the board's need
Electrolyzer oxygen	H&sb2; plant	spirulina pond + greenhouse aeration	21 kt/yr
Electrolyzer hydrogen	H&sb2; plant	ammonia → the GCT fertilizer chain	2.63 kt/yr
Electrolyzer reject water / heat / flush	H&sb2; plant	product line / desal preheat / brine line	nothing leaves
DC reject heat	compute park	the same seawater pipe, arriving warm at the membranes	−$0.007/$m^3$
Kiln + acid exotherm	board kilns + metals plant	desal preheat + greenhouses	30 GWh/yr
CO&sb2; (kiln + power)	combustion streams	mineralized into board (2 Mt PG) · pond carbon	540 t CO&sb2;/yr to the ponds
Phosphogypsum	GCT stack	fire-rated board — with its retardant from our own brine	2 Mt/yr
Acid stream	GCT wet-process acid	U + heavy REE extraction; raffinate back to fertilizer	716.5k lbs U&sb3;O&sb8;/yr
Municipal return flows	city mesh	30%+ reuse · pump-as-turbines on the descent	3.9 GWh/yr
Sludge / digestate	wastewater + organics	biogas for kilns and greenhouses	booked option, honestly labeled

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Slaughterhouse bones	local slaughterhouses	bone char → the fluoride filters; **spent char → phosphate soil amendment for the fields**	filter loop closes into fertilizer
Spent pond medium	spirulina harvest	nutrient-rich → greenhouse fertigation	nothing discarded
Ammonia fertilizer	GCT chain	the terfas fields' host plants + greenhouse soils	the truffle grows on the loop's own nitrogen
Compute services	colo park	export — the data product	tenant-gated
Ra-226 / Po-210	acid stream radiology	stays in the gypsum/board matrix — the radiology gate	below the 1,000 Bq/kg exemption
Board offcuts	board line	recycled into new board	zero waste
Sodium chloride	brine line	industrial offtake at the park gate	nothing hauled
Halophytes / Artemia on brine	brine line	future option, honestly labeled — not a current credit	the only stream awaiting its consumer

the ponds drink the desal permeate — ~{:.0f} m³/d, {:.2f}% of the 150,000 m³/d output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the ~30 g/L ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

3. The capital stack and the ask

Phase 0: $150-250k, 90 days, no construction — the price of replacing every projection with a measurement (itemized in the complete audit). **Then, module by module, on public gates:** $58k health pilot · $100M metals · $550M solar · $165M desal · $5-30M truffles · $2-5M food. Security per module: offtake assignment, STEG PPA, municipal contracts, MIGA/DFI channel — modeled, not committed.

What moves "commercially validated" from zero

GCT MOU + first assay batch · STEG PPA award · desal LOI ≥60% at ≥$0.68/m³ · first truffle fruit + ≥€20/kg contract · 100-point water panel + 90-day health KPIs · 3,000-family cohort baseline. Each is a Phase-0/1 deliverable with a date.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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